

Summarizer

〈POWrangers〉
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POW!

01

Threads



02

**Bad
Thread?**



03

Game



04

**Code
exercise**



Thread

Allows a program to operate more efficiently by doing multiple things at the same time



ruinedchildhood
imgur.com/q9o6q

Threads

- Can be used to perform complicated tasks in the background without interrupting the main program;
- Have access to the same variables, allowing them to share data;



Creating a Thread



```
public class Main implements Runnable {  
  
    public void run () {  
        System.out.println("Hey girl, are you a Java Compiler?" +  
            "Cause when I proposed a date"+  
            "you said I wasn't your type");  
    }  
}
```

**Implementing the
Runnable Interface**

VS

**Extending the
Thread Class**

```
public class Main extends Thread {  
  
    public void run () {  
        System.out.println("Why do blind programmers use Java?" +  
            "Because they can't C");  
    }  
}
```



BAD THREAD!!!!

SOLVING A PROBLEM WITH MULTI-
THREADING?



YOU NOW **Three** PROBLEMS. HAVE

Tapar source lalalala

Visibility

Variables are not always read or written from memory but from cache.

Atomicity

Everything Volatile, when increments create a new object plus "one". (?)

Ordering

Java compiler can change the order of execution if it believes the outcome will not change.

Locks!!



04

code exercise

Before I run out of time



Concurrent Web server

```
public class Main {  
    public static void main(String[] args) {  
  
        ServerSocket serverSocket = new ServerSocket (8989);  
  
        while (true){  
            Socket request = serverSocket.accept();  
            Thread thread = new Thread (new RunServer(request));  
            thread.start();  
        }  
    }  
}
```

```
public class RunServer implements Runnable {  
    Socket request;  
    public RunServer(Socket request) {  
        this.request = request;  
    }  
}
```

```
@Override
public void run(){
    System.out.println(Thread.currentThread().getName() + " Thread");

    BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(request.getInputStream()));

    BufferedWriter out = new BufferedWriter(new OutputStreamWriter(request.getOutputStream()));

    File file = new File("resources/HTML.html");

    String header = "HTTP/1.0 200 Document Follows\r\n" +
        "Content-Type: text/html; charset=UTF-8\r\n" +
        "Content-Length: " + file.length() + " \r\n" +
        "\r\n";

    String holder = bufferedReader.readLine().split(" ")[1];
    String[] arrayHtml = bufferedReader.readLine().split(" ");

    System.out.println(arrayHtml[1]);
    out.write(header);

    FileReader fileReader = new FileReader(file);
    BufferedReader bufferFileReader = new BufferedReader(fileReader);

    String temp;
    while ((temp = bufferFileReader.readLine()) != null) {
        out.write(temp);
    }
    out.close();
}
```

Concurrent Web server

```
org.bootcamp.powrangers.Main x
/Library/Java/JavaVirtualMachines/openjdk-8.jdk/Contents/Home/bin/java ...
Thread-0 Thread
localhost:4020
Thread-1 Thread
localhost:4020
Thread-2 Thread
localhost:4020
Thread-3 Thread
localhost:4020
Thread-4 Thread
```



**AWESOME
Game!!**

A method with a name, parameter types, and return type, but without an implementation.

Compiler

Constructor

Overloading

Void

Overriding

Binary Search

Abstract Method

Concatenation

Interface

**Occurs when more than one meaning
is assigned to a method name.**

Compiler

Constructor

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Interface

A method that is automatically called and runned when an object is created by a class.

Compiler

Constructor

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Interface

A fast algorithm to find a value in a sorted array. It narrows the search down to half of the array in every step.

Compiler

Constructor

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Interface

Added to a method when nothing is returned.

Compiler

Constructor

Overloading

Void

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Concatenation

Interface

Redefining a method in a subclass.

Compiler

Constructor

Overloading

Void

Overriding

Binary Search

Abstract Method

Concatenation

Interface

**A reference type, that can contain only constants,
method signatures and nested types.**

Compiler

Constructor

Overloading

Void

Overriding

Binary Search

Abstract Method

Concatenation

Interface

A program that translates code in a higher-level language to a machine instructor.

Compiler

Constructor

Overloading

Void

Overriding

Binary Search

Abstract Method

Concatenation

Interface

Placing one string after another to form a new string .

Compiler

Constructor

Overloading

Void

Overriding

Binary Search

Abstract Method

Concatenation

Interface



BOOM!!!!

Thank you!

